# Reproducibility and Good Practices in Science

Pietro Franceschi | pietro.franceschi@fmach.it

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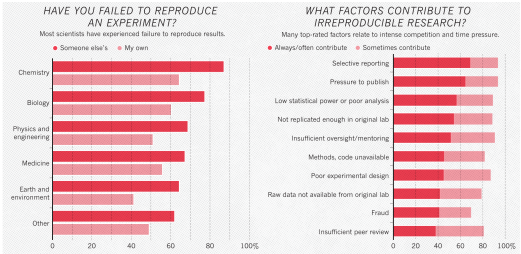
# Why does this matter?

Science is the best tool we have to understand the natural world.

But science is done by **humans**, and humans make mistakes — or worse.

The question is not *whether* errors happen, but **whether we can detect and correct them**.

# The Scientific Method




Baker, M. (2016). 1,500 scientists lift the lid on reproducibility. *Nature*, 533, 452–454. Survey of **1,576 researchers**.

The cornerstone of science is **reproducibility**: an independent group must be able to repeat your experiment and obtain the same result.

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- **Poor experimental design** → systematic bias
- **Inadequate statistical analysis** → spurious findings
- **Data analysis errors** → wrong conclusions from correct data
- **Confirmation bias** → seeing what we want to see
- **Fraud and misconduct** → active manipulation

# A spectrum of failure

**The "reproducibility crisis"**

A large fraction of published scientific results cannot be reproduced by independent groups. Estimates range from **50% to 70%** of findings in some fields.

We will look at **three real cases** spanning different failure modes:

1. Honest delusion — Benveniste & homeopathy (Nature, 1988)
2. Fatal errors — The Potti scandal at Duke University
3. Correlation is not causation — Chocolate and Nobel Prizes (NEJM, 2012)

Section 1

# Case 1: The Benveniste Affair

# The original claim

In **June 1988**, *Nature* published a paper by Dayenas, Beauvais, and **J. Benveniste** from INSERM 200, Paris.

The claim: human basophils degranulate in response to anti-IgE antibodies diluted to $\mathbf{10^{-120}}$ — far beyond Avogadro's limit.

Not a single antibody molecule could be present in the solution.



SCIENTIFIC PAPER

NATURE VOL. 333 30 JUNE 1988

## Human basophil degranulation triggered by very dilute antiserum against IgE

E. Dayenas, F. Beauvais, J. Amara*, M. Oberbaum*, B. Robinzon †, A. Miadonna ‡, A. Tedeschi, B. Pomeranz§, P. Fortier§, P. Belon, J. Sainte-Laudy, B. Poitevin & J. Benveniste§

INSERM U 200, Université Paris-Sud, 32 rue des Carnets, 92140 Clamart, France
* Department of Medical & Clinical Immunology, Kaplan Hospital, Rehovot 76100, Israel
† Department of Animal Sciences, Faculty of Agriculture, PO Box 12, The Hebrew University of Jerusalem, Rehovot 76100, Israel
‡ Department of Internal Medicine, Infectious Diseases and Immunopathology, University of Milano, Ospedale Maggiore Policlinico, Milano, Italy
§ Departments of Zoology and Physiology,* Ramsay Wright Zoological Laboratories, University of Toronto, 25 Harbord Street, Toronto, Ontario, Canada M5S 1A1
To whom correspondence should be addressed.

*When, human polymorphonuclear basophils, a type of white blood cell with antibodies of the immunoglobulin E (IgE) type on its surface, are exposed to anti-IgE antibodies, they release histamine from their intracellular granules and change their staining properties. We report here experiments in which histamine release, assayed by a change in the basophils' staining properties, was measured as a dependent variable from 40 to 60% of the basophils, despite the calculated absence of any anti-IgE molecules at the highest dilutions. Since dilutions need to be accompanied by vigorous shaking for the effects to be observed, transmission of the biological information could be related to the molecular organization of water.*

THE antibodies responsible for human immediate hypersensitivity belong to the IgE isotype<sup>1</sup>. The mast cell/basophil membrane through receptors with high affinity<sup>2</sup>. Cross-linking of membrane IgE molecules by a polyvalent antigenic stimuli such as allergens or anti-IgE antiserum that can bridge IgE molecules in membrane. This process triggers transmembrane and intracellular signals followed by granule exocytosis with the release of histamine and loss of metachromasia of basophil granules stained with a mixture of toluidine blue. Optical basophil degranulation is well correlated with other *in vivo* and *in vitro* procedures for the diagnosis of allergy<sup>3-5</sup>.

In preliminary experiments, degranulation of human basophils contained in leukocyte suspensions was induced not only by the usual concentration of anti-IgE antibody (1 x $10^4$ dilution in the assay) but also by dilutions of 1 x $10^{40}$ to 1 x $10^{80}$ of the antibody in the assay, but also by very low concentrations of this antibody (2.2 x $10^{-103}$ M), where the number of IgG antibody molecules in the assay is supposedly too low to trigger the process. We thus further explored this phenomenon.

Rabbit polyclonal anti-human IgE antibodies (Fc) antiserum (1 mg specific antibody per ml) were prepared in HEPES-buffered Tyrode's solution containing human serum albumin (USA) down to 1 x $10^{18}$ dilution, corresponding to a 2.2 x $10^{-96}$ M theoretical concentration (th) in the assay (see Fig. 1 legend for methods). The expected basophil degranulation, which was assessed by counting cells with metachromatical loss of granules, was found to remain stable from the undiluted serum to low antiserum dilutions with a maximum at ~ 1 x $10^2$ dilution. Successive peaks of degranulation varying between 40 and 80% were then found down to 1 x $10^{40}$ dilution, with periods of 1 x $10^{20}$ dilution (Fig. 1a). The specificity of the observed effects at high dilutions was serially diluted a hundred-fold down to 1 x $10^{80}$ (to give 2.2 x $10^{-103}$ M th in the assay) and similar results were obtained (Fig. 1b). Degranulation induced by high dilutions of anti-IgE antiserum was observed in ten experiments on the full range of dilutions down to 1 x $10^{40}$ when at least 70 similar results were obtained at one or the other part of the high dilution range (see Table 1 for statistical analysis of the 70 experimental results). As controls, goat antihuman IgG (Fc) antiserum (Fig. 1b, n = 4) or Tyrode's solution containing HSA (n = 5) were diluted down to 1 x $10^{20}$ and 1 x $10^{4}$, respectively. Cells incubated in conditions identical to those with anti-IgE antiserum dilutions but containing no IgE (n = 3) remained stable. Anti-IgE-induced degranulation were reproducible, but the peaks of degranulation could shift by one or two dilutions with every fresh sequential dilution of anti-IgE and dependent on the blood sample. The waves of basophil degranulation were also seen with substances other than anti-IgE anti-serum at high and low dilutions, such as monoclonal anti-human IgE antibodies, anti-IgD antibodies, the calcium ionophore A23187 (calcimycin), phosphalipase A2 from bee venom or porcine pancreas, the $Na^+$ ionophore monensin (up to 90% degranulation at 1 x $10^{-6}$ M th) and the $Ca^{++}$ ionophores A23187 and ionomycin (1 x $10^{-6}$ M th). The specificity of the observed effects at high dilutions was assessed by the following experiments. Solutions filled with antiserum against IgE$\lambda$ was further unfraught dissolved in the ionophore experiments, because removing the corresponding ion from the cellular environment blunted basophil

**Table 1** Basophil counts after exposure to anti-IgE antiserum at low and high dilutions

| Samples | Experiment 1 | Experiment 2 | Experiment 3 | Experiment 4 |
|---|---|---|---|---|
| Tyrode's+HSA* | 81.0±1.4 | 82.5±1.5 | 81.5±1.5 | 82.0±1.4 |
| Tyrode's+HSA | 81.0±1.4 | 81.7±1.4 | 81.0±1.4 | 80.0±1.1 |
| Tyrode's+HSA | 80.0±1.0 | 80.0±2.3 | 81.7±1.8 | 80.7±0.9 |
| $a[1 \times 10^{2}]$ | 77.0±0.8 (1.000) | 77.0±0.8 (1.000) | 77.0±0.8 (1.000) | 77.0±0.8 (1.000) |
| $a[1 \times 10^{4}]$ | 77.0±0.8 (1.4) | 69.3±1.2 (.3) | 66.3±2.2 (.04) | 70.7±0.8 (.02) |
| $a[1 \times 10^{6}]$ | 78.0±1.1 (.4) | 68.7±1.0 (.1) | 72.7±1.4 (.4) | 74.7±2.0 (.06) |
| $a[1 \times 10^{8}]$ | 77.0±0.8 (1.000) | 77.0±0.8 (1.000) | 77.0±0.8 (1.000) | 77.0±0.8 (1.000) |
| $a[1 \times 10^{20}]$ | 45.0±0.5 (.000) | 75.0±1.4 (.001) | 41.3±1.8 (.000) | 48.3±2.2 (.000) |
| $a[1 \times 10^{40}]$ | 49.0±1.7 (.000) | 50.3±0.5 (.000) | 51.0±2.5 (.032) | 78.3±2.2 (.28) |
| $a[1 \times 10^{60}]$ | 79.0±1.0 (.3) | 69.0±0.7 (.1) | 75.3±1.7 (.9) | 69.3±0.7 (.1) |

Blind experiments: test tubes were randomly coded twice by two independent pairs of observers and assayed. The codes were simultaneously broken at the end of all experiments. Each value is the mean ± s.e.m. of basophil number actually counted in triplicate (see legend to Fig. 1 for methods. % Number in parentheses indicates percentage degranulation compared with Tyrode's+HSA.

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This would be the **scientific basis for homeopathy**.

*Nature* published it **anyway** — with an unprecedented *editorial reservation*, and on the condition that an independent team could investigate the laboratory.

# Nature sends an investigation team

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One week on-site. Access to **laboratory notebooks**. They imposed **strictly blinded** protocols.

# The result: a delusion



NATURE, VOL. 334 28 JULY 1988 — NEWS AND VIEWS — 287

## "High-dilution" experiments a delusion

*The now celebrated report by Dr J. Benveniste and colleagues elsewhere is found, by a visiting Nature team, to be an insubstantial basis for the claims made for them.*

THE remarkable claims made in Nature (*Nature*, 333, 816; 1988) by Dr Jacques Benveniste and his associates are based chiefly on an extensive series of experiments which are described in detail in their paper. We know of no substantial effort has been made to exclude systematic error, including observer bias, and whose interpretation has been clouded by the exclusion of certain observations. We are not alone in that anti-IgE at "high dilution" will deregranulate basophils. The phenomenon described is not reproducible in the ordinary meaning of that word.

We conclude that there is no substantial basis for the claim that anti-IgE at high dilution retains its biological activity. We remain its biological effectiveness, and that the hypothesis that water can be imprinted with the memory of past solutes is an unnecessary as it is fanciful.

We are not saying that the effects—reportedly, these solutions contain no molecules of anti-IgE, and so are not solutions in the ordinary sense. "Solute-free solution" would similarly be illogical.

We have been investigating this long visit to Dr Benveniste's laboratory, the INSERM unit for immunopharmacology and allergy (otherwise INSERM 200) at Clamart, in the western suburbs of Paris, from 10 to 16 July. We arrived on the 10th. Among other things, we were dismayed to learn that the salaries of two of Dr Benveniste's coauthors of the published article are paid for under a contract between INSERM and Laboratoires Jacques Benveniste SA, a supplier of pharmaceuticals and homoeopathic medicines, as were our hotel bills.

Benveniste's results are being widely publicized as a challenge to conventional medicine. In the light of our investigation, we believe that such use amounts to misuse.

Our visit and investigation were undertaken at the behest of Nature, the original article. We acknowledge that we are an oddly constituted group. One of us (J.R.) is a professional magician (and also a MacArthur Foundation fellow) whose skills are often employed to detect fraud in case the remarkable results reported had been produced by trickery. Another of us (W.W.S.) has been chiefly concerned, during the past decade, in studies of scientific fraud, particularly in the scientific literature, and with the behavioural misconduct in science. The third (J.M.) is a journalist with a background in theoretical physics. None of us has first-hand experience in the field of work at INSERM 200.

We acknowledge that we might well have misunderstood what we were shown of the work of the laboratory. But, on the basis of our experience, we are confident that the design of the experiments reported by INSERM 200 is inadequate as a basis for the claims made last month and that the defects are shed catalogue are a sufficient explanation of the remarkable results then reported.

We believe that experimental data have been unartistically assessed and that imperfections inadequately reported. We found that Dr Benveniste is a charismatic and then cherished a delusion about the interpretation of its data.

We are grateful to Dr Jacques Benveniste for his openness in discussing his experimental results and for the facilities allowed us to borrow and to photocopy the relevant laboratory notebooks, which were invaluable for our investigation. We have every reason to believe that Dr Benveniste and his colleagues are convinced of the reality of the phenomena reported in his article. We are also in the debt of several of Dr Benveniste's colleagues, especially to Dr Elisabeth Davenas, for their generosity in the matter of demonstrating the standard ideation experiments and of repeating them in a blinded protocol under our scrutiny. We know that our report will be a disappointment to many of them. We are sorry.

What follows is a narrative account of our visit and a summary of our conclusions.

Our investigations concentrated mainly on the experiments on basophil degranulation which the publication was based. During our week in Paris, we resisted several proffered opportunities to examine other systems in which high dilution is claimed



Fig. 1 *A demonstration degranulation, the first of the three open experiments.*

not to diminish the biological effectiveness of a molecule.

The experimental system has evolved from a test for assessing the susceptibility of human blood basophils to the releasing principle A first found from allergens to have the specific effect of interacting with the leukocytes known as basophils, causing them to degranulate — that is, to release the contents of cytoplasmic granules carrying histamine and other substances which mediate the symptoms of asthma and hay-fever.

These allergic reactions are apparently mediated at least in part by IgE molecules attached to the surfaces of basophils (in the sense of the word "attached" above). Normally, degranulation is triggered by the interaction of anchored IgE molecules with an antigen, but the same effect can be brought about by the use of anti-IgE—a solution of molecules that bind to human IgE into an animal of another species. (INSERM 200 uses goat anti-IgG at a concentration of 1 mg cm$^{-1}$ sold by the Dutch company Nordic.)

The laboratory's notebooks provide ample evidence that this expected degranulation is a maximum between log(dilution) 2 and 4.

Benveniste described the published experiments. In the standard procedure, a buffered solution of anti-IgE is serially diluted by a factor of 10 by transferring measured volumes from one test-tube to another. Pipette tips are discarded after each transfer. Human blood samples are suspended while cells drained from human blood are transferred to wells in a polystyrene plate. To each of these is added a measured volume of serially diluted anti-IgE solution. The plates' wells are incubated for 30 minutes at 37°C. An acidic solution of toluidine blue, which stains intact but not degranulated basophils red, is added and the numbers of stained basophils counted. A saline or a buffered solution of anti-IgE, which does not degranulate basophils, is used as a control.

We were surprised to learn that the experiments described in the paper (*Nature*, 333, 816; 1988) have been periods of several months at a time during which solutions at high dilution have not degranulated basophils. Indeed, the laboratory had just emerged from such a period. Speculation at the laboratory is that the chemicals or reagents have been contaminated or otherwise made unsuitable.) It also appears that

*Nature* Vol. 334, **28 July 1988**

The investigators concluded:

> *"We conclude that there is no substantial basis for the claim that anti-IgE at high dilution retains its biological effectiveness."*

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- **Financial conflict of interest** — two co-authors were paid by a homeopathic company

# The lesson from Benveniste

**Extraordinary claims require extraordinary evidence**

Blinding, independent replication, and rigorous documentation are not bureaucratic obstacles — they are the defences against our own wishful thinking.

Benveniste was probably not lying — he genuinely believed his results.

That is precisely what makes this case so instructive.

Section 2

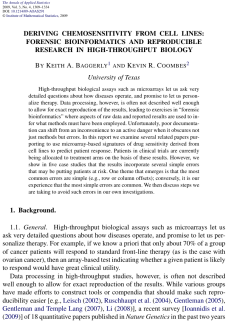
# Case 2: The Potti Scandal

# The promise: personalised cancer therapy

Starting in **2006**, Anil Potti and colleagues at **Duke University** published a series of high-profile papers.

The claim: microarray **gene expression signatures** from cancer cell lines can predict which drug a patient will respond to.

Patients were being **enrolled in clinical trials** and allocated to treatment arms on this basis.



The Annals of Applied Statistics
2009, Vol. 3, No. 4, 1309–1334
DOI: 10.1214/09-AOAS291
© Institute of Mathematical Statistics, 2009

**DERIVING CHEMOSENSITIVITY FROM CELL LINES: FORENSIC BIOINFORMATICS AND REPRODUCIBLE RESEARCH IN HIGH-THROUGHPUT BIOLOGY**

BY KEITH A. BAGGERLY[1] AND KEVIN R. COOMBES[2]
*University of Texas*

High-throughput biological assays such as microarrays let us ask very detailed questions about how diseases operate, and promise to let us personalize therapy. Data processing, however, is often not described well enough to allow for exact reproduction of the results, leading to exercises in "forensic bioinformatics" where aspects of raw data and reported results are used to infer what methods must have been employed. Unfortunately, poor documentation can shift from an inconvenience to an active danger when it obscures not just methods but errors. In this report we examine several related papers purporting to use microarray-based signatures of drug sensitivity derived from cell lines to predict patient response. Patients in clinical trials are currently being allocated to treatment arms on the basis of these results. However, we show in five case studies that the results incorporate several simple errors that may be putting patients at risk. One theme that emerges is that the most common errors are simple (e.g., row or column offsets); conversely, it is our experience that the most simple errors are common. We then discuss steps we are taking to avoid such errors in our own investigations.

**1. Background.**

1.1. *General.* High-throughput biological assays such as microarrays let us ask very detailed questions about how diseases operate, and promise to let us personalize therapy. For example, if we know a priori that only about 70% of a group of cancer patients will respond to standard front-line therapy (as is the case with ovarian cancer), then an array-based test indicating whether a given patient is likely to respond would have great clinical utility.

Data processing in high-throughput studies, however, is often not described well enough to allow for exact reproduction of the results. While various groups have made efforts to construct tools or compendia that should make such reproducibility easier [e.g., Leisch (2002), Ruschhaupt et al. (2004), Gentleman (2005), Gentleman and Temple Lang (2007), Li (2008)], a recent survey [Ioannidis et al. (2009)] of 18 quantitative papers published in *Nature Genetics* in the past two years

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[1]Supported in part by NIH Grant P50 CA083639 09.
[2]Supported in part by NIH Grant P50 CA070907 08.
*Key words and phrases.* Microarrays, reproducibility, forensic bioinformatics.

# The forensic bioinformatics

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> *"Data processing is often not described well enough to allow for exact reproduction of the results, leading to exercises in **forensic bioinformatics**."*
> — Baggerly & Coombes, *Ann. Appl. Stat.* 2009

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The most common errors are simple errors.

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**This was not fraud in the traditional sense**

The errors were "simple" — but they were never caught because the raw data and analysis code were never shared.

# The lesson from Potti

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  *"Poor documentation can shift from an inconvenience to an active danger when it obscures not just methods but errors." — Baggerly & Coombes*

Section 3

# Case 3: Chocolate and Nobel Prizes

# Published in the New England Journal of Medicine

In **October 2012**, Franz Messerli published in *NEJM* (one of the most prestigious medical journals):

**"Chocolate Consumption, Cognitive Function, and Nobel Laureates"**

The logic: flavonoids in chocolate improve cognitive function → countries that eat more chocolate should have more Nobel laureates.

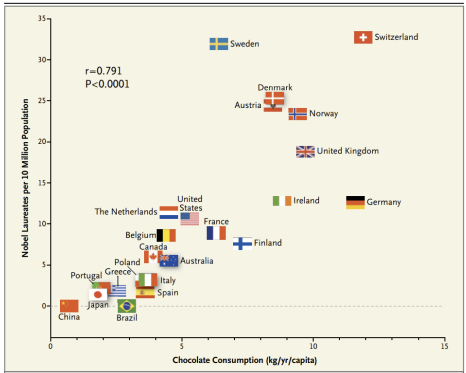



Figure 1. **Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.**

# The statistics

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To increase a country's Nobel laureate count by **1**, it would need to increase chocolate consumption by **0.4 kg per capita per year**.

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$$r = 0.791, \quad P < 0.0001$$

Highly statistically significant.

To increase a country's Nobel laureate count by **1**, it would need to increase chocolate consumption by **0.4 kg per capita per year**.

The paper **concludes**: *"Chocolate consumption enhances cognitive function, which is a sine qua non for winning the Nobel Prize."*

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**The lesson**

the prestige of a journal does not guarantee the validity of a finding. **Critical reading is always required.**

# The lesson from chocolate & Nobel prizes

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- With enough variables, spurious correlations are inevitable
- Always ask: *What is the plausible mechanism? What are the confounders?*

Section 4

# Three Cases, Common Threads

# What connects them?

| Case | Main failure |
|---|---|
| Benveniste | Confirmation bias, lack of blinding |
| Potti | Undocumented analysis, no data sharing |
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All three were eventually corrected — but only after **harm was done**.

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- **Replicate** before acting on findings, especially high-stakes ones

# Take-home message

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It is the mechanism by which science corrects itself — and the guarantee that our findings are worth building on.

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**In metabolomics**, where data matrices are large, analysis pipelines are complex, and the pressure to find biomarkers is high, these principles are not optional.